

§ Modules / 模

Idea F field, $V : F$ -vector space is a group (additive), $(V, +)$ endowed with $(\cdot) : F \times V \rightarrow V$ s.t.

$$t(v_1 + v_2) = tv_1 + tv_2, \quad (t_1 + t_2)v = t_1v + t_2v, \quad 1_F v = v.$$

Clearly: can replace F by any ring.

Definition R ring, a (left) **R -module** is an additive group $(M, +)$ endowed with $(\cdot) : R \times M \rightarrow M$ s.t.

$$r(x + y) = rx + ry, \quad (r_1 + r_2)x = r_1x + r_2x, \quad (r_1r_2)x = r_1(r_2x), \quad 1_R x = x$$

where $r, r_1, r_2 \in R, x, y \in M$. Shorthand: M .

The right R -module is similar except for $x(r_1r_2) = (xr_1)r_2$.

Trivial R -module: $\{0\}$.

Remark Left R -module = right R^{op} -module where R^{op} is the ring with the same $(R, +)$ but $x \odot y = yx$. Hence we shall focus on left modules. Note that R commutative $\implies$ no difference!

General properties

- $0_R x = 0_M$,
- $(-1_R)x = -x$,
- $(n \cdot 1_R)x = nx, \forall n \in \mathbb{Z}$.

Examples

- F field, then F -modules are F -vector spaces.
- R is a R -module where $+, \cdot$ are those in R .
- Take $R = \mathbb{Z}$, $\mathbb{Z}$ -module = an additive ring $(A, +)$ with $\mathbb{Z} \times A \rightarrow A$ + properties.

But: $\forall n \in \mathbb{Z}, \underbrace{na}_{\text{module}} = \underbrace{na}_{(A,+)} \implies$ the module structure is determined by $(A, +)$.

Conversely: given $(A, +)$, setting $\mathbb{Z} \times A \rightarrow A : (n, a) \mapsto \underbrace{na}_{(A,+)}$ makes A into a $\mathbb{Z}$ -module.

Hence $\mathbb{Z}$ -modules = additive groups (= abelian groups), and the notation of homomorphisms coincide.

Definition Let M_1, M_2 be R -modules. A map $f : M_1 \rightarrow M_2$ is said to be a homomorphism if

- f is a homomorphism of abelian groups,
- $f(rx) = rf(x), \forall r \in R, x \in M_1$.

$$\text{Hom}(M_1, M_2) = \text{Hom}_R(M_1, M_2) = \{f : M_1 \rightarrow M_2 \mid \text{homomorphisms}\}.$$

$\text{Hom}(M_1, M_2)$ is an additive group via $(f_1 + f_2)(x) = f_1(x) + f_2(x)$.

Caution In general, $\text{Hom}(M_1, M_2)$ is not an R -module by setting $(rf)(x) := rf(x)!$
 $\because (rf)(r'x) = rf(r'x) = rr'f(x) \neq r'r f(x) = r'((rf)(x))$. However if R commutative, then $\text{Hom}(M_1, M_2)$ is an R -module by setting $(rf)(x) := rf(x)$.

Homomorphisms are closed under composition. Composition is “bi-additive”:

$$(f_1 + f_2)g = f_1g + f_2g, \quad f(g_1 + g_2) = fg_1 + fg_2.$$

Definition Isomorphisms from M to M' means a homomorphism $f : M \rightarrow M'$ s.t.

$\exists g : M' \rightarrow M$ homomorphism, $gf = \text{id}_M, fg = \text{id}_{M'}$. Such g is unique (the inverse map of f). We say M is isomorphic to M' if $\exists f : M \xrightarrow{\sim} M'$.

Notation $f : M \xrightarrow{\sim} M', M \simeq M'$.

Easy properties

- Closed under composition,
- id_M is a $\simeq$,
- $f : \simeq \implies f^{-1} : \simeq$.

Proposition Let $f : M \rightarrow M'$ homomorphism, if f is bijective, then f is $\simeq$.

Proof Same as groups / vector spaces. It boils down to show: the inverse map $f^{-1} : M' \rightarrow M$ is a homomorphism. $\square$

Definition Submodule $N \subset M$ is a subset that is a subgroup under $+$ and closed under multiplication by R . N is an R -module itself, and the inclusion $N \hookrightarrow M$ is a homomorphism.

Examples

- R field $\implies$ get vector subspaces. $R = \mathbb{Z} \implies$ subgroups.
 R commutative, $M = R \implies$ submodules = ideals of R .
- Let $(M_i)_{i \in I}$ be a family of submodules of $M \implies \bigcap_{i \in I} M_i$ is a submodule of M .
- Define $\sum_{i \in I} M_i := \left\{ \text{finite sums } \sum_{i \in I} x_i \mid \forall i \in I, x_i \in M_i \right\}$, it is a submodule of M .

It is also the smallest submodule of M that contains all M_i .

- S any subset of M , set $\langle S \rangle := \left\{ \text{finite sums } \sum_{s \in S} r_s s \mid r_s \in R \right\} = \sum_{s \in S} R s$.

It is the smallest submodule of M containing every element in S .

$$\langle s_1, \dots, s_n \rangle := \langle \{s_1, \dots, s_n\} \rangle.$$

Observation Let $f : M \rightarrow M'$ be a homomorphism, $\ker(f) := f^{-1}(0_{M'})$, $\text{im}(f)$ = its image, both are submodules (of M and M' , respectively), $\because f(rx) = rf(x)$.

§ Quotient modules

R ring, $M : R$ -module, $N \subset M$ submodule. Cosets of N in $M :=$ subsets of the form $x + N$ in M .

Goal Upgrade $(M/N, +)$ to an R -module s.t. the quotient map $q : M \rightarrow M/N$ is a homomorphism of modules.

Definition-Proposition On the quotient group $(M/N, +)$ define

$$\begin{aligned} (\cdot) : R \times M/N &\longrightarrow M/N \\ (r, x + N) &\mapsto rx + N \end{aligned}$$

This makes M/N into an R -module s.t. q is a homomorphism of modules. We call M/N the quotient module of M by N .

Proof Well-defined: suppose $x + N = x' + N$, then $\exists u \in N$ s.t. $x' = x + u$, then

$$rx' + N = rx + ru + N = rx + N.$$

Associativity / distributivity and other axioms: reduce to the level of M .

Finally, q is already a homomorphism of abelian groups. Also

$$q(rx) = rx + N = r(x + N) = rq(x).$$

$\square$

This is also the unique module structure on M/N that makes q R -linear.

R field $\longrightarrow$ quotient spaces, $R = \mathbb{Z} \longrightarrow$ quotient groups.

Definition Let $f : M \rightarrow M'$ homomorphism, its **cokernel** $:= \text{coker}(f) := M' / \text{im}(f)$.

f surjective $\iff \text{im}(f) = M' \iff \text{coker}(f) = \{0\}$. Also f injective $\iff \ker(f) = \{0\}$.

Proposition (universal property of quotients) Let $f : M \rightarrow M'$ homomorphism, $N \subset M$ submodule, $N \subset \ker(f)$. Then $\exists!$ homomorphism $\bar{f} : M/N \rightarrow M'$ s.t. $f = \bar{f}q$ and $\text{im}(f) = \text{im}(\bar{f})$.

Proof Reduce to the level of additive groups. Remains to show: $\bar{f}$ is an R -module homomorphism.

$$\bar{f}(rx + N) = f(rx) = rf(x) = r\bar{f}(x + N).$$

□

Corollary Given $f : M \rightarrow M'$ homomorphism, $N \subset M, N' \subset M'$ submodules, $f(N) \subset N'$, then $\exists! \bar{f} : M/N \rightarrow M'/N'$ s.t. $q'f = \bar{f}q$.

Proof Take $f = f'q'$ in previous Prop.

□

Proposition $f : M \rightarrow M'$ homomorphism, then $\bar{f} : M/\ker(f) \xrightarrow{\sim} \text{im}(f) \subset M'$.

Proof Known that $M/\ker(f) \xrightarrow{\bar{f}} \text{im}(f)$ as groups. But $\bar{f}$ is a homomorphism of R -modules, hence it is $\xrightarrow{\sim}$ of R -modules.

□

Proposition $f : M \rightarrow M'$ surjective homomorphism, then:

$$\begin{aligned} \{\text{submod } N' \subset M'\} &\xleftrightarrow{1:1} \{\text{submod } \ker(f) \subset N \subset M\} \\ N &\longmapsto f^{-1}(N') \\ f(N) &\longleftarrow N \end{aligned}$$

preserving $\subset$, and if $N' \longleftrightarrow N$ then the composite of $M \rightarrow M' \rightarrow M'/N'$ induces $M/N \xrightarrow{\sim} M'/N'$.

Proposition M R -module, $M, N \subset M$ submodules $\implies M/(M \cap N) \xrightarrow{\sim} (M + N)/N$.

Proof Upgrade from the group version.

□